MealMate — UML Diagrams

Visual representation of the MealMate system using standard UML notation, rendered with Mermaid.js.

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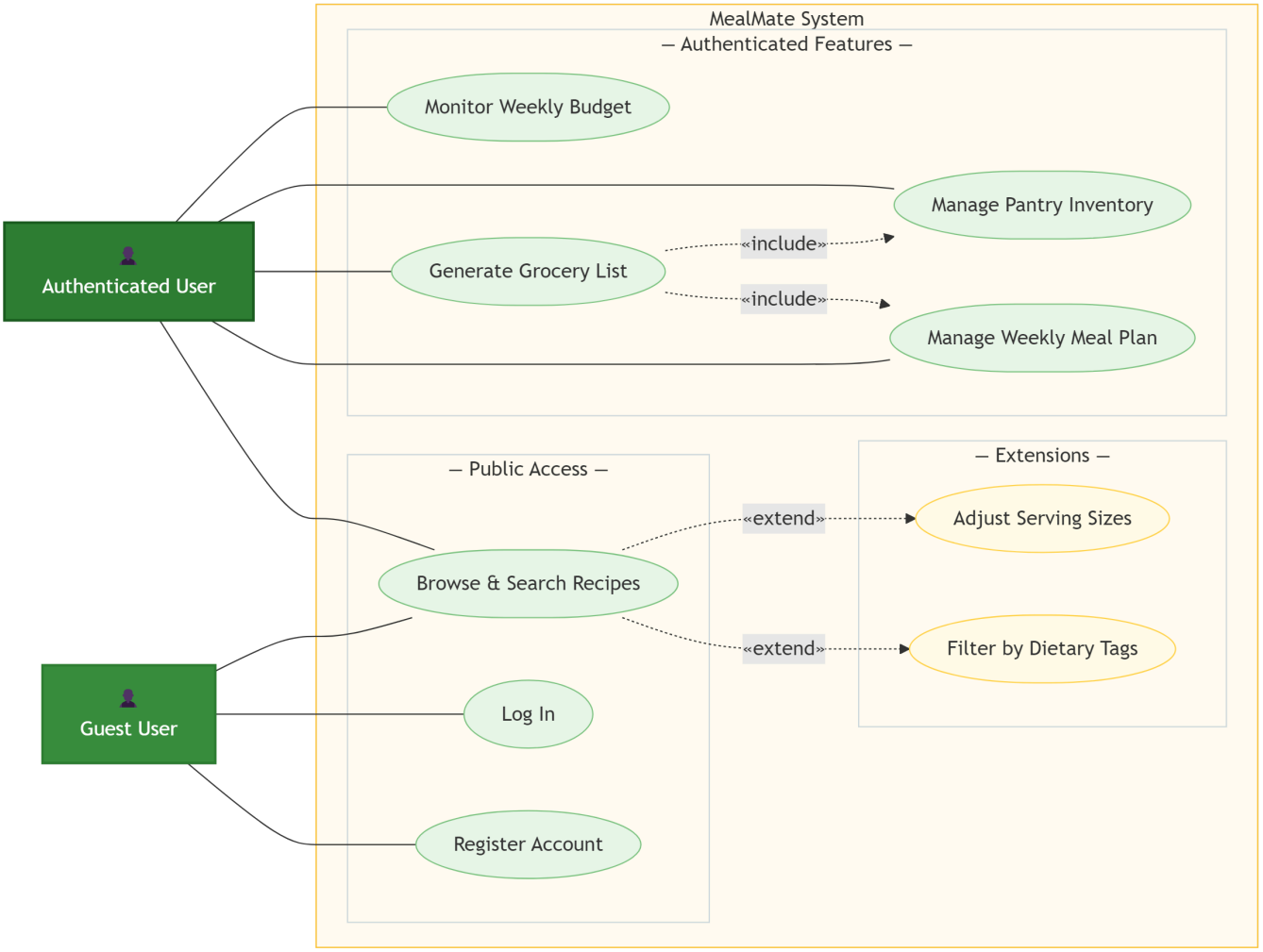
1. Use Case Diagram



> 💡 The use-case model distinguishes between **Guest** and **Authenticated** users. Public access includes browsing recipes, registration, and login. Authenticated users can additionally manage meal plans, pantry inventory, grocery lists, and weekly budgets. **«include»** arrows denote mandatory sub-flows (e.g. generating a Grocery List always includes the Meal Plan and Pantry), while **«extend»** arrows denote optional behaviour (e.g. Browse Recipes can be extended with dietary filtering or serving adjustments).

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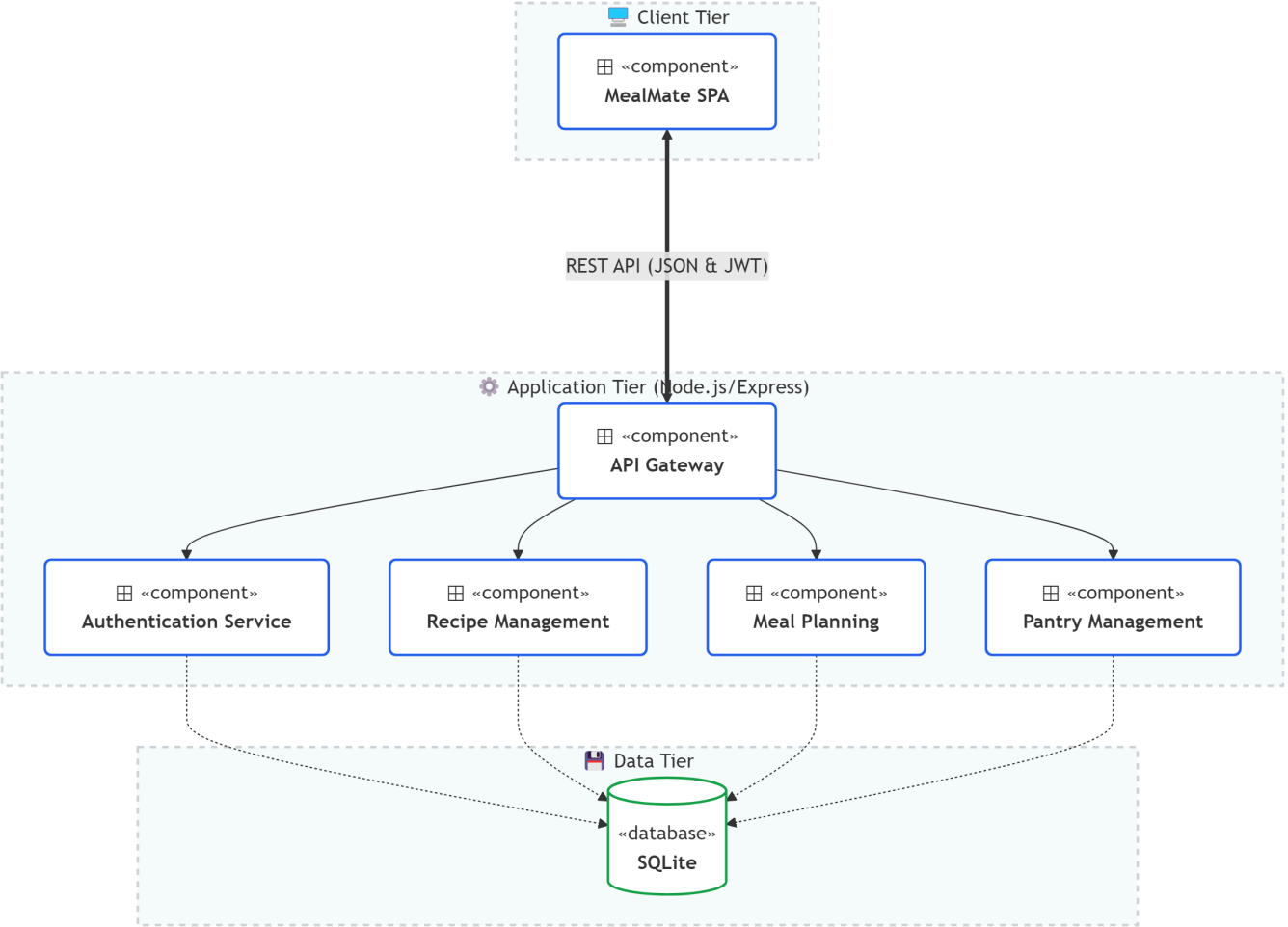
2. Component Diagram



> 💡 MealMate follows a classic **Client-Server** pattern. The **React + Vite** frontend runs entirely in the browser, managing state through an `AuthContext` that persists the JWT in `localStorage`. Every protected API call attaches the token as a `Bearer` header. The **Node.js / Express** backend exposes four REST route groups (`/auth`, `/recipes`, `/mealplans`, `/pantry`), all backed by a single **SQLite** file.

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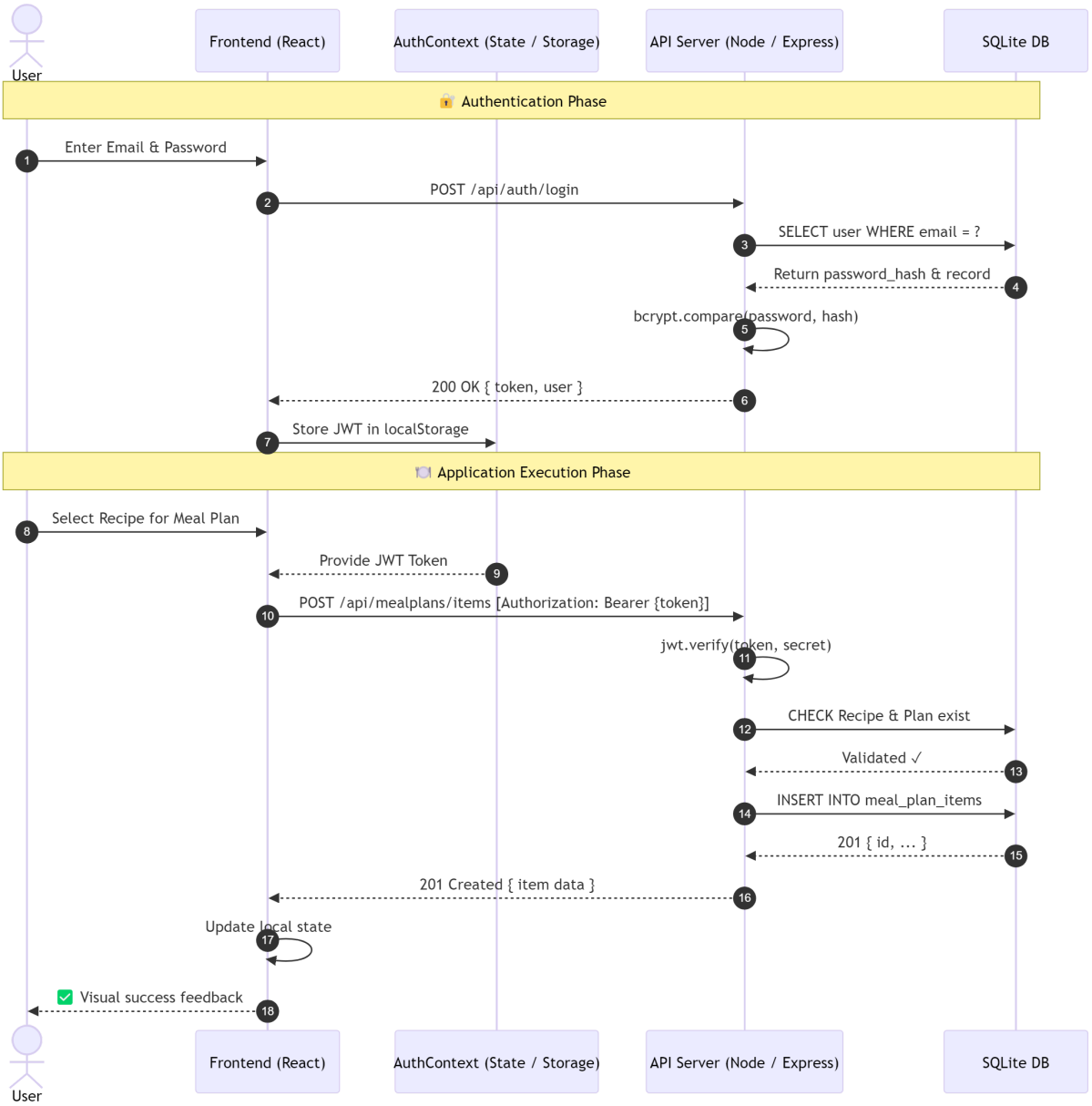
3. Sequence Diagram



> 💡 The flow is split into two phases. In the **Authentication Phase** the frontend POSTs credentials, the server verifies the password hash with `bcrypt`, and returns a signed JWT that is stored in `localStorage`. In the **Application Execution Phase** the stored token is attached to subsequent requests; the server validates the signature with `jwt.verify()` before writing to the database, ensuring only authenticated users can modify meal plan data.

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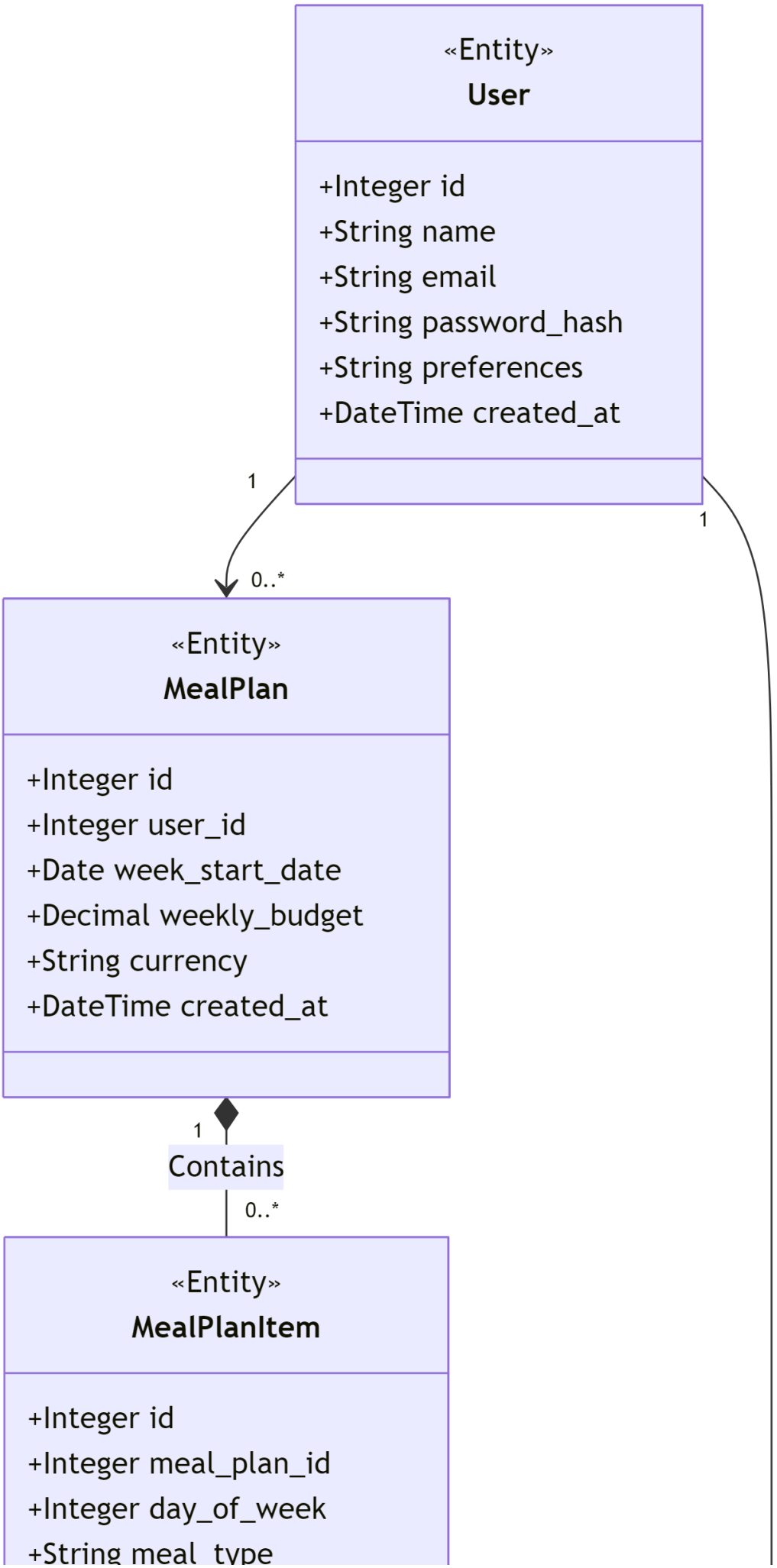
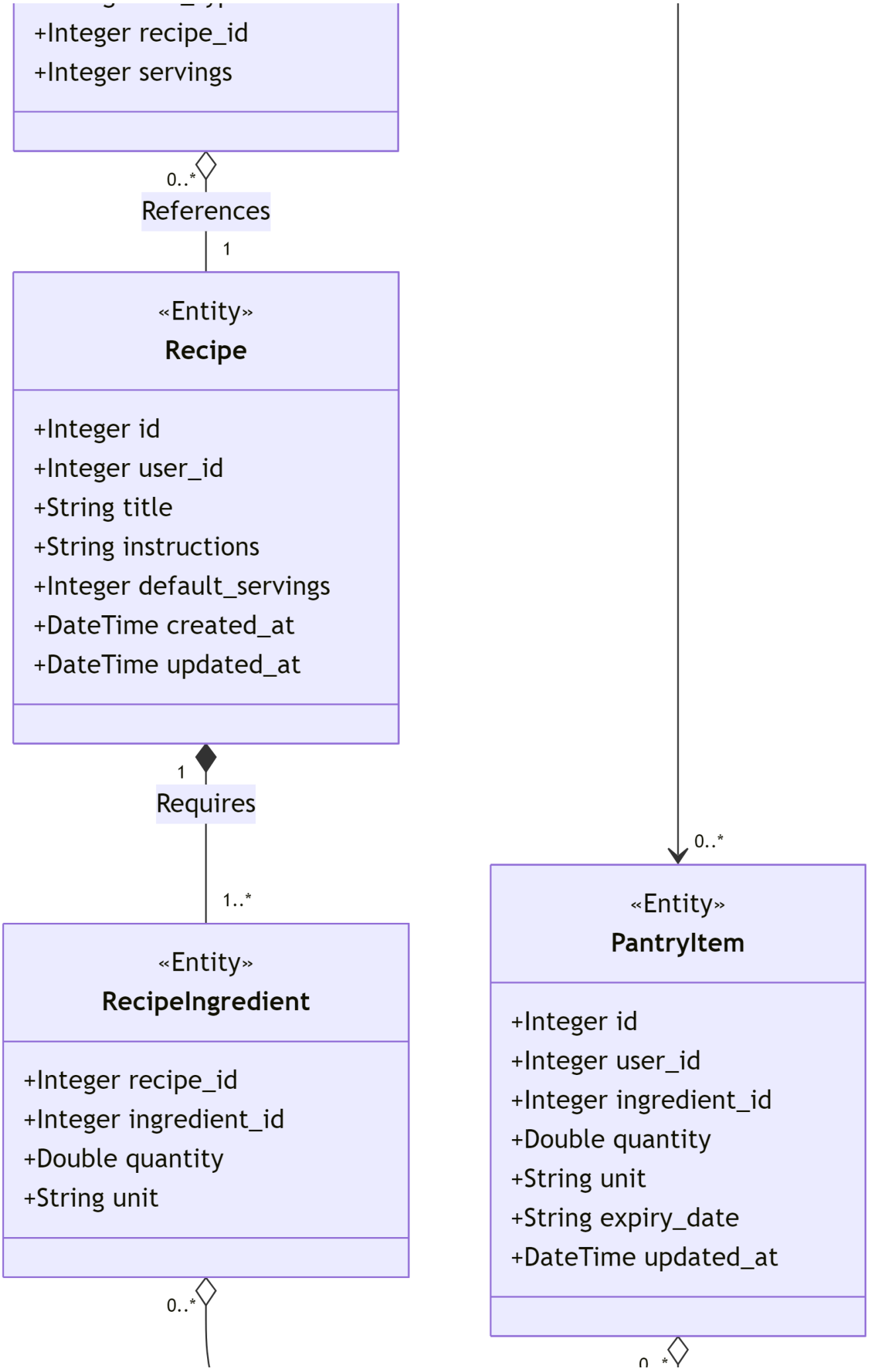
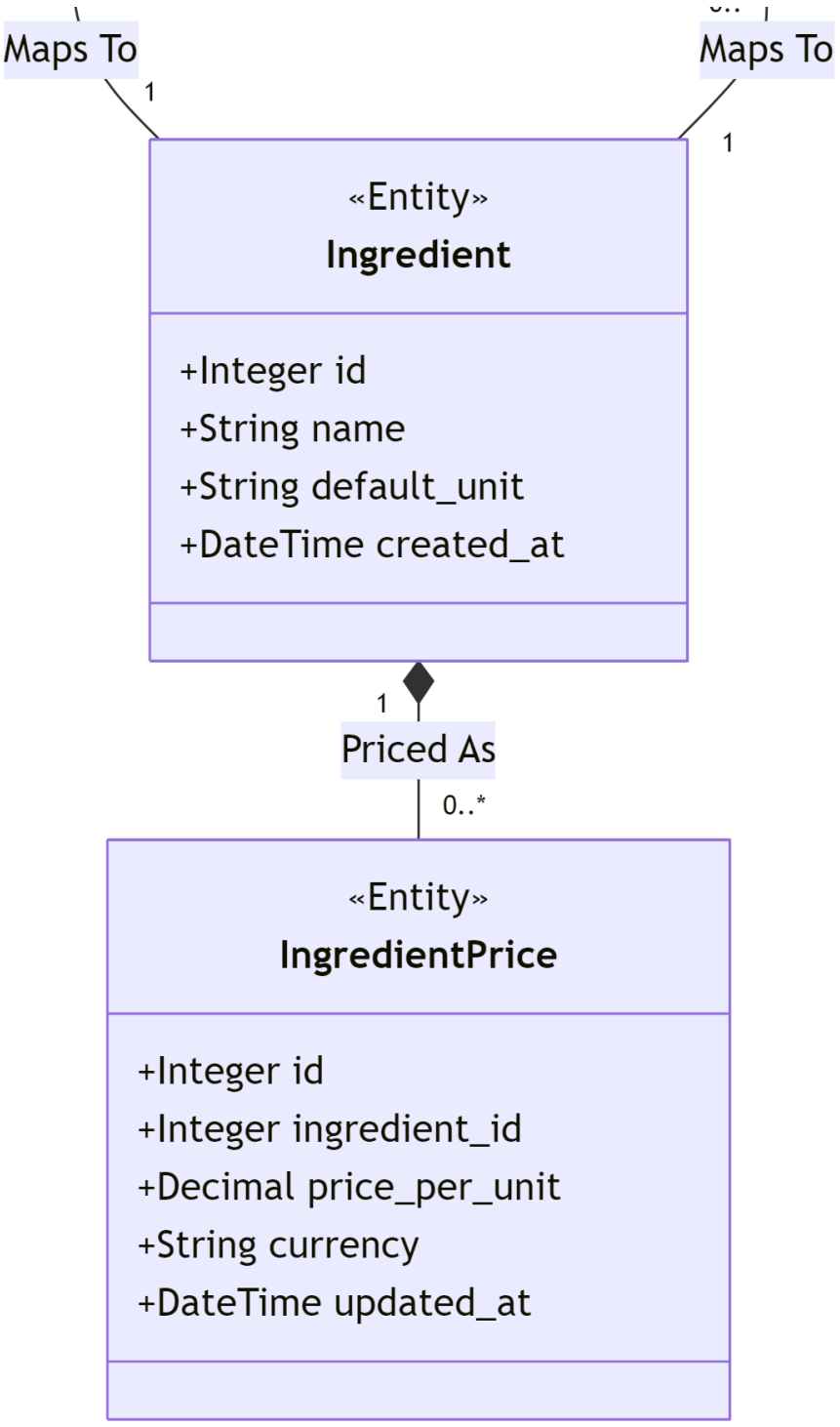
4. Class Diagram


«Entity»
User
+Integer id
+String name
+String email
+String password_hash
+String preferences
+DateTime created_at
1
0..*
1
«Entity»
MealPlan
+Integer id
+Integer user_id
+Date week_start_date
+Decimal weekly_budget
+String currency
+DateTime created_at
1
Contains
0..*
«Entity»
MealPlanItem
+Integer id
+Integer meal_plan_id
+Integer day_of_week
+String meal_type


+Integer recipe_id
+Integer servings
0..*
References
1
«Entity»
Recipe
+Integer id
+Integer user_id
+String title
+String instructions
+Integer default_servings
+DateTime created_at
+DateTime updated_at
1
Requires
1..*
«Entity»
RecipeIngredient
+Integer recipe_id
+Integer ingredient_id
+Double quantity
+String unit
0..*
0..*
«Entity»
PantryItem
+Integer id
+Integer user_id
+Integer ingredient_id
+Double quantity
+String unit
+String expiry_date
+DateTime updated_at



> 💡 Every **User** owns zero-or-more **MealPlan** and **PantryItem** records. A `MealPlan` is composed of `MealPlanItem` rows (one per meal slot), each referencing a **Recipe**. Recipes are built from `RecipeIngredient` join records that map to shared **Ingredient** entities — keeping ingredient names canonical, and can be categorized using `RecipeTag` and `Tag` entities. Each `Ingredient` has zero-or-more **IngredientPrice** entries used for budget calculation. `PantryItem` links a user's stock to the same `Ingredient` catalogue.